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Research Proposal

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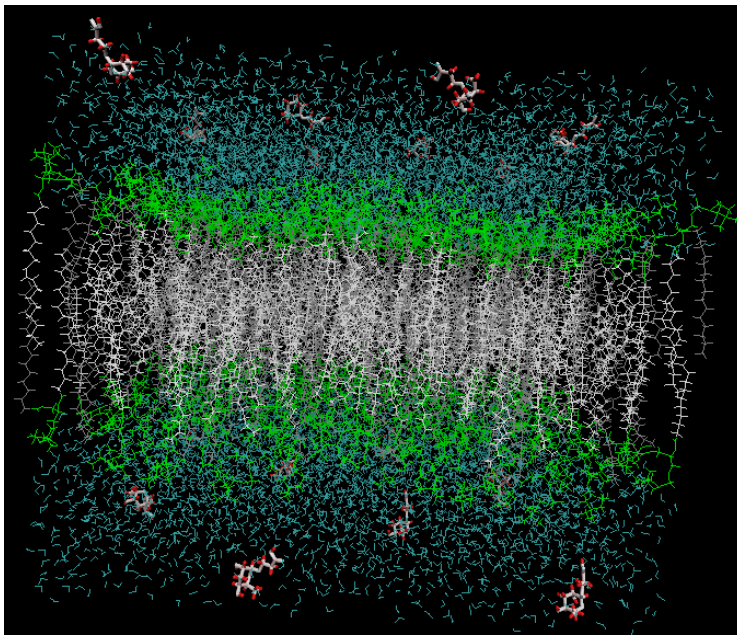

Simulation of C3 ligands in POPC membrane visualized using VMD

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Prediction of Chemical Ligands Using Molecular Dynamic Simulations

Objectives:

- Determine how Molecular Dynamics (MD) simulations can improve the homology models and identify stable ligand structures
- Use molecular dynamic simulations to predict the ligand that bind to the receptors *in silico*

Background:

For this project, we will first be focusing on receptors that sense pheromones in C. *elegans* before modelling on other organisms. Majority of these chemoreceptors are G-Protein Coupled Receptors (GPCRs). An overview of several ligand discovery studies show that GPCR models, despite structural inaccuracies, can be efficiently used to find novel ligands for GPCRs. [1]
Each organism typically encodes ~1000 different chemoreceptors and which are very large in size. Thus, prediction of which chemical ligand a given organism will respond to, by just looking at genome sequences, is impossible because the number and size of data creates a complex combinatorial problem to identify ligands. Another problem that arises is that chemoreceptors rapidly evolve by modifying their ligand specificity and activity. Thus, lessons learned from one species cannot be applicable to others.
Our goal is to overcome these problems and predict if an organism can respond to a given chemical ligand. Our approach for this was to first get protein homology model for discovered gold standards and then use it for MD simulations to predict the ligand for receptors *in silico*. This can bypass the need for using experimentally challenging methods. It can also be modelled for many different organisms. The gold standards used are the 4 receptors in C. *elegans* that sense C3: *srg-34*, *srg-36*, *srg-37* and *CBG24690*. [2]

Summary of previous work:

Once we obtained the different homology models for proteins, we had to find a way to insert these proteins into lipid membranes and then run simulations on them. Initially, we used CHARMM-GUI, which provides a web-based graphical user interface to generate molecular simulation systems [3], to insert the protein into a DOPC lipid membrane and generate a protein data bank (PDB) file of the entire system. This was then prepped with the AMBER preparatory program [4] and then used for MD simulation with AMBER.
At first, the membranes kept breaking apart as shown in Figure 1. Several test runs were done to optimize the parameters for a simulation using AMBER MD and values for keeping the membrane stable were found. The final pipeline to run the complete simulation was determined to be as shown in Figure 2.

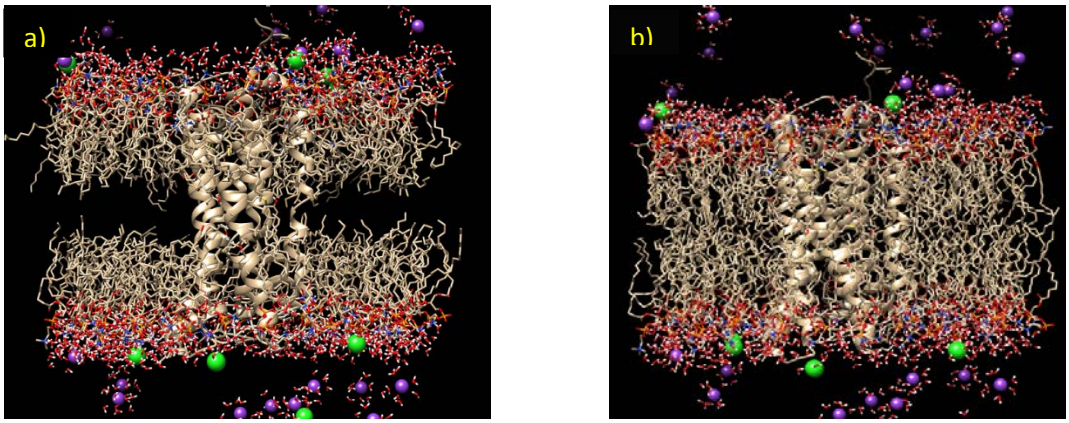

Figure 1) Final simulation of srg-34 inserted in DOPC lipid bilayer before (a) and after (b) optimizing parameters

We also ran simulations of the ligands both alone and in the membrane to see how the ligands will interact with the membranes.

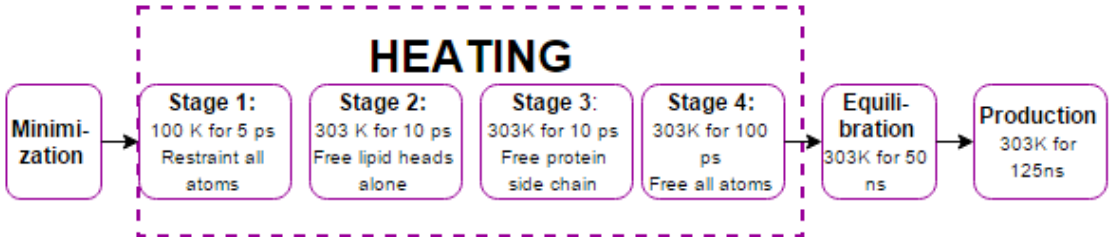

Figure 2) Pipeline to run simulation of protein in membrane

Once each part worked individually, we decided to integrate all the steps and write a wrapper script so that all these steps can be automated. For this, we had to shift from a web based GUI and build the membrane in command line. This was done using VMD, a molecular visualization program for displaying and analyzing large biomolecular systems using 3-D graphics.[5] We wrote a custom script that takes as input the size of the membrane needed and generates a POPC membrane of given size. Another script was written to insert 'n' number of ligands in random positions and sample MD simulations were ran.
The next step was to insert the protein into the membrane from command-line. For this, we used CHIMERA's [6] match maker program to align a new protein to a manually aligned protein in a membrane and discover the coordinates of the new protein. This pipeline needs to be further edited to include the preparatory steps in Amber and run a simulation.

Methods:

Running automated simulation of protein and ligand in lipid membrane:
The next step is to write a wrapper script that adds both the protein and ligands into the membrane and run simulations. The script should also prep the files and start to run simulations. The final pipeline should look like as shown in Figure 3.

The main parts of this wrapper script are:

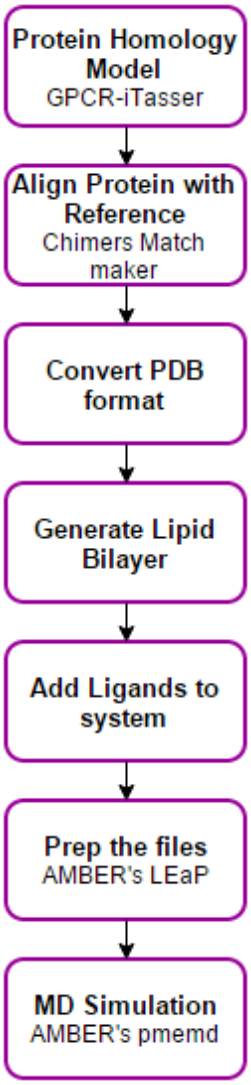

Figure 3) Main pipeline for wrapper script

- **Protein Homology Model**

The protein homology models we are using for this project were got using GPCR-iTasser. [7]

- **Align with reference**

One of the challenging tasks of inserting a protein into a lipid membrane is to figure out the orientation of the protein in the membrane. Doing this for each simulation manually would be time consuming. So, we use one system as reference, where the protein was manually aligned in the membrane, and align new protein files to it to generate coordinates for it. This can be done on the command line using CHIMERA's python library that has a module for aligning proteins called "Match Maker". [6] Once this has been done, some trimming of the C-terminus might be needed to ensure that excess of the tail end is not still in the lipid membrane.

- **Convert format**

The naming conventions used for the PDB file generated by GPCR-iTasser are slightly different from the ones that AMBER uses. To make the file compatible with amber, we need to convert the format.

- **Generate lipid layer**

Once we have a compatible PDB file, we need to generate the lipid membrane around the protein. This can be done using VMD. Once the membrane is generated, we need to remove any conflicting molecules to allow for proper insertion. A script to insert the proteins into the membrane and removing conflicting molecules has already been written.

- **Add Ligands**

To this system, we should be able to add ligands. Previously, for testing purposes, we had used random orientations for the ligands. Now, we will use AutoDock Vina [8] to dock the ligands and identify their most likely positions. Using energy of binding, 3 of the best poses will be used to start the simulation process.

- **Prep files for simulation**

Once the PDB file of the system is ready, we can use AMBER's preparatory programs LEaP to prep the files for simulation. This step will help us generate two files, namely the parameter file and the topology file. The former includes parameters necessary for setting up energy calculations while the later assigns charges to each atom present in the file.

- **MD simulation**

The final step here is to run the MD simulation using AMBER's programs SANDER and PMEMD. The parameters to run these have already been determined by us previously.

Analyzing the simulation:

Once the simulations are done, we also need to come up with different analysis methods.

To determine the accuracy of the simulation of the protein membrane, we used a script to determine the "Area per lipid" of the system. Typical to other lipid simulations, the area per lipid is seen to converge after sometime for our test runs of the membrane.

To keep track of the movement of the ligands, we wrote a script that calculates the z-axis value of the ligand over the entire simulation run.

The main thing we need to look into now is Root Mean Square Deviation (RMSD) profiles. RMSD measures the deviation of a target set of coordinates to a reference set of coordinates. This can be calculated by:

$$\text{RMSD} = \sqrt{\{\sum_{i=0}^{N}[\text{m}(i) * \left(X(i) - Y(i)\right)^2]\} / \text{M}}$$

Where N is the number of atoms, m(i) is the mass of atom i, X(i) is the coordinate vector for target atom i, Y(i) is the coordinate vector for the reference atom i, and M is the total mass.

We need to calculate the RMSD values for each transmembrane domain, inner loop and outer loop in the GPCR and also compare the position and orientation of each ligand.

Conclusion:

The survival of animals largely depends on their ability to sense their external environment using chemoreceptors. There is general interest in understanding their function, as they can be used to manipulate sensory pathways to combat parasites and pests. This project helps us bypass experimentally challenging methods to study the interaction of ligands to GPCRs by studying them *in silico*. Last semester was spent testing and optimizing values for the simulations and setting up scripts to help automate whole pipeline. This semester, our main goal is to use the tested values to run batch simulations and analyze the results efficiently.

Reference:

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